

Circuit Analysis (Lab # 2)

Experiment 2: Measurements of Waveforms and Fundamental Quantities, Basic resistive circuits.

Name:

Registration No:

Date:

Grade and Signature:

CLO3	Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objectives: This experiment will provide exposure to measurements of waveforms and Fundamental Quantities, Basic resistive circuits. A primary purpose of this lab course is for you to master the use of electronic test equipment and measurement of fundamental quantities.

Equipment required:

Power Supply, Breadboard, Oscilloscope, Signal Generator Connecting wires, Resistances, Multimeter.

Procedure

Experiment # 1

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'A' shown below in Figure 1.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 100 Hz and peak-to-peak voltage of 5V.
4. Set up one probe across input signal, and another across R1 and R2 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

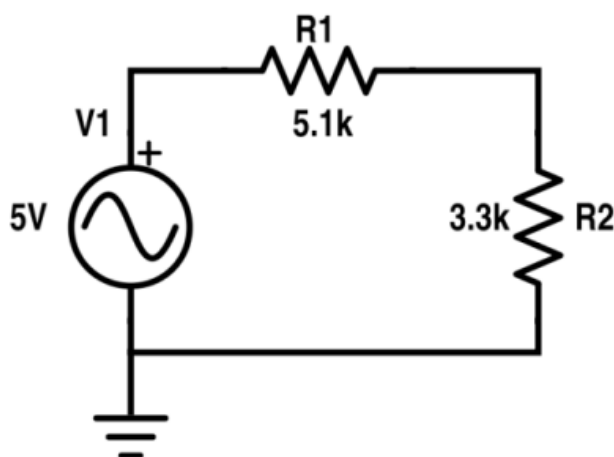


Figure 1 Circuit 'A'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1			
R2			

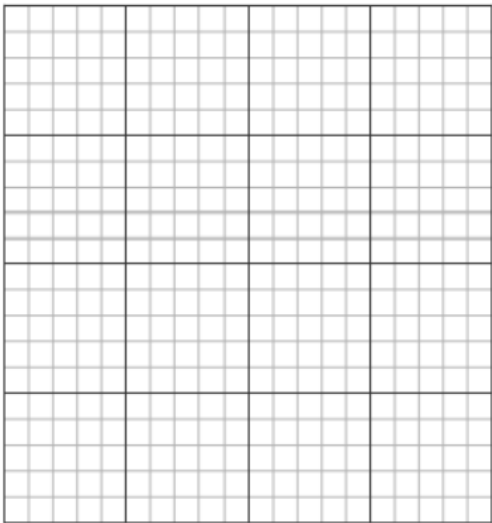


Figure 2 Input Voltage Signal

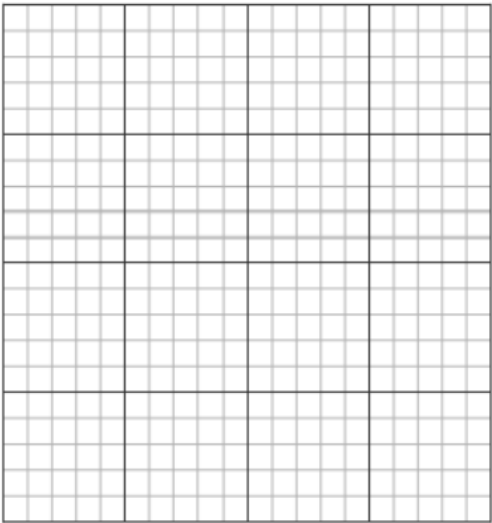


Figure 3 Voltage Signal across R1

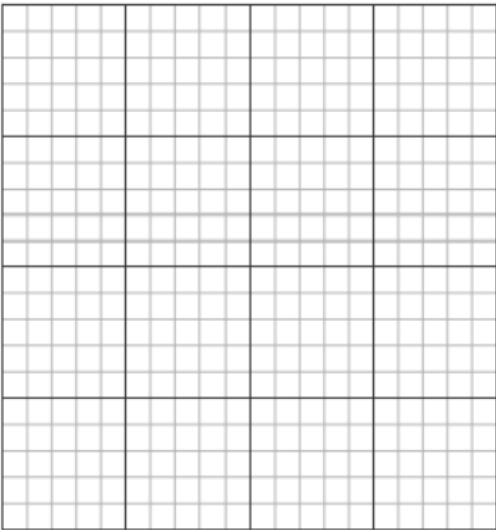


Figure 5 Voltage Signal across R2

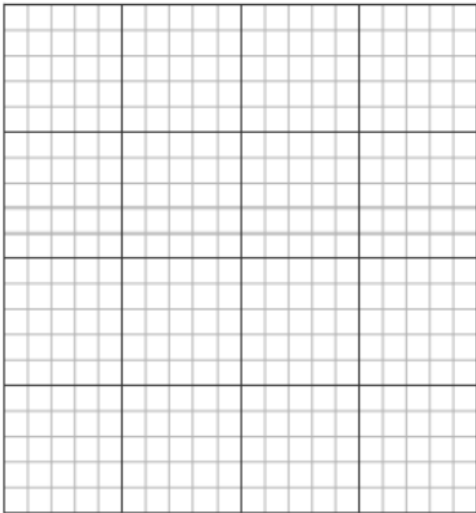


Figure 6 Draw 7V Sinusoidal

Experiment # 2

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'B' shown below in Figure 7.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 200 Hz and peak-to-peak voltage of 7V.
4. Set up one probe across input signal, and another across R1, R2 and R3 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

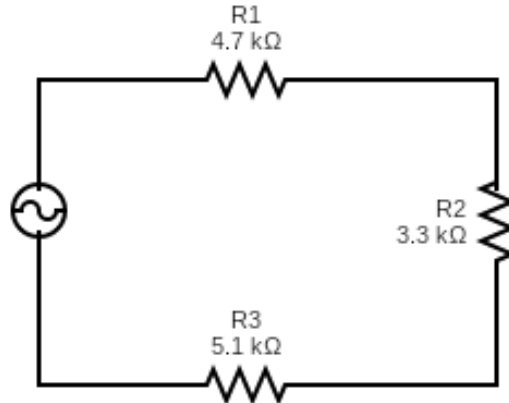


Figure 7 Circuit 'B'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1			
R2			
R3			
R1 and R2			
R2 and R3			

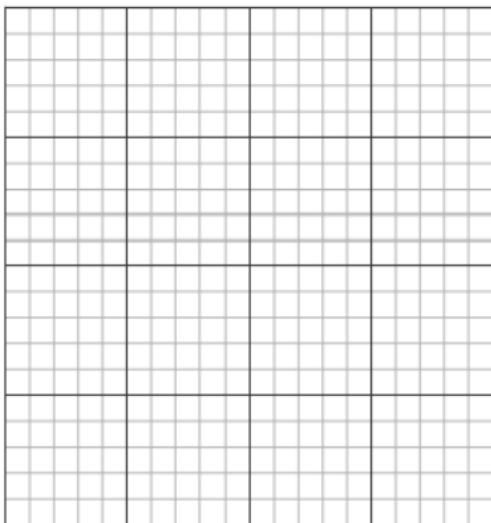


Figure 8 Input Voltage Signal

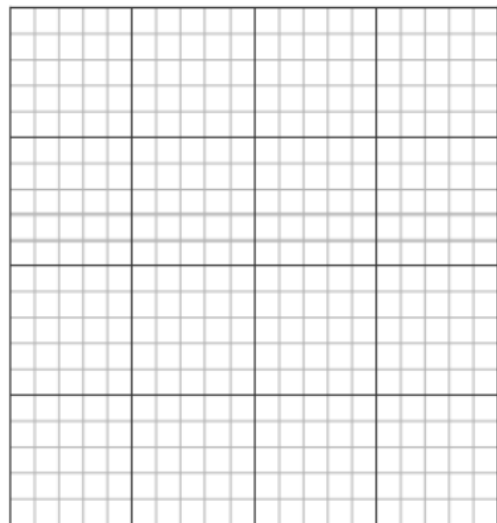


Figure 9 Voltage Signal across R1

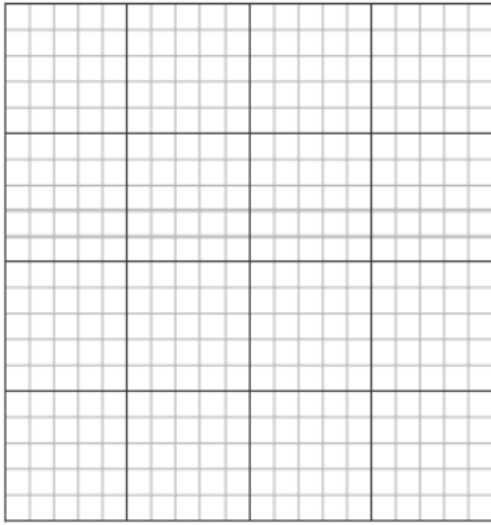


Figure 10 Voltage Signal across R2

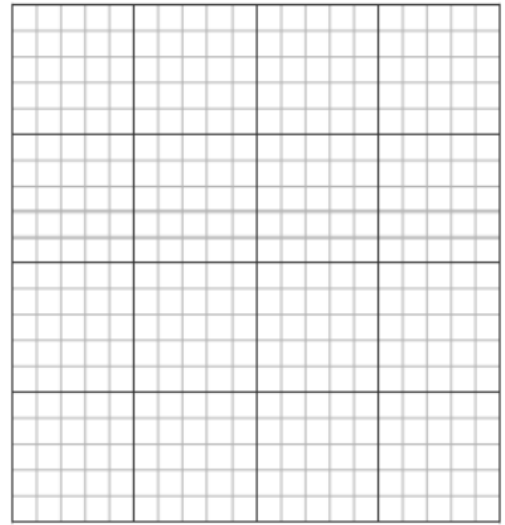


Figure 11 Voltage Signal across R3

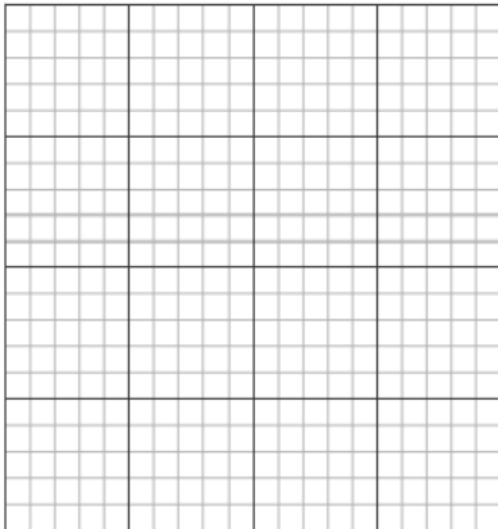


Figure 12 Voltage Signal across R1 and R2

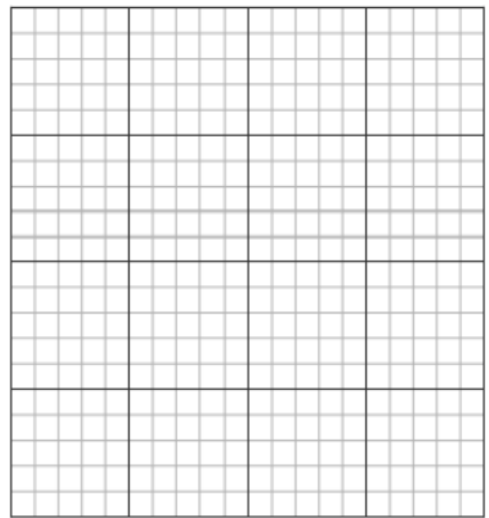


Figure 13 Voltage Signal across R2 and R3

Post Lab Questions

1. Define Ohm's law.

2. What is resistor?

3. How resistor works?

4. What is your inference from the input and output waveforms on oscilloscope?

5. Show calculations of 7V input signal (Figure 10) from signal generator.
 - a. Voltage calculations (Following volt/dev and number of boxes of oscilloscope)
 - b. Frequency/Period calculations (Following Steps in frequency measurements section)
 - c. Current calculation (Ohm Law)

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